

SHOULD-COST ANALYSIS REPORT

Generated: 13 Aug 2026 · 11:26

Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Spur Gear, m3 z38, 30mm face, bored

Commodity: GEAR · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 4 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥193.14	60.3%	14.8%	4.6%	7.4%

Model Confidence: **Low** · 4 traced operations · 10 data points auditable

Engine Warnings (4)

Nitriding holds the as-cut geometry (class 7), so no hard finishing was added — this is the reason the route is specified. Confirm the flank load case suits a thin nitrided case rather than a deep carburised one.

88 of 88 gear shop parameters are representative values, not verified data (source set: "representative (no plant data supplied)"). Cutting speeds, feeds, tool life and heat-treat rates were not taken from a standard, a machine-builder catalogue or your plant. This estimate shows the right structure and is NOT a quotable number until the shop data is replaced.

Gear route: Gear hobbing -> Chamfer and deburr -> Nitride (gas/plasma) -> Gear inspection — cutting cycle 1.88 min/part; perishable tooling £0.23/part; NRE £18500 amortised over 200,000 parts.

Raw Material bucket = blank material £3.84 + nitride (gas/plasma) £8.77/part ($2.088 \text{ kg} \times £4.2/\text{kg} = £8.77$ (furnace, by weight — a 10-90 h cycle, so dearer per kg than carburising; the saving is the grinding it avoids)) — a purchased service bought by weight, so it carries no machine hours.

Blank turning and all tooth cutting/finishing are separate operations in the process bucket.

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 266.0 cm³; mass for the selected material family 2.088 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: gear

Alloy / material: 4140 / 42CrMo4 / EN19 (Chromoly Alloy Steel) · Net weight: 2.088 kg (measured)

Material utilisation: 100% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 4

\$1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥116.44	60.3%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
2. Process (Machine)	¥28.49	14.8%	
3. Direct Labour	¥8.92	4.6%	
4. Tooling (amortised)	¥2.90	1.5%	
5. Packaging	¥1.18	0.6%	
6. Logistics	¥2.08	1.1%	
Factory Cost	¥160.02	82.9%	
7. Overhead (SG&A)	¥18.81	9.7%	
Subtotal	¥178.83	92.6%	
8. Supplier Margin	¥14.31	7.4%	
TOTAL SHOULD-COST	¥193.14	100.0%	

§2 — Commercial Parameters

Overhead Rate	12.0%	Supplier Margin	8.0%
Packaging / part	¥1.18	Logistics / part	¥2.08
Total Tooling Cost	¥580461.77	Amortisation Volume	200,000 parts

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-steel4140	ID	
Grade / Specification	4140 / 42CrMo4 / EN19 (Chromoly Alloy Steel)		UK steel stockholder, Jun 2026. Index-anchored 2026-07 refresh. Regional adj. ×0.880 (millSteel)
Region	China		
Net Finished Weight	2.0880 kg	kg	Weight in finished part
Direct Material Cost	¥35.46	CNY	Bypasses weight-based model

Data Confidence	Medium
Effective Date	2026-07

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Blank turning (face, OD, bore)	CNC Lathe (2-axis)	¥185.20	4.93	1	80.0%	¥19.04
Gear hobbing	CNC Gear Hobber — small ("d m 4 , "d Ø 2 0 0)	¥265.03	1.16	1	80.0%	¥6.40
Chamfer and deburr	Gear Chamfer / Deburr Cell	¥123.89	0.43	1	80.0%	¥1.10
Gear inspection	CNC Gear Checker (profile / lead / pitch)	¥272.86	0.34	1	80.0%	¥1.95
TOTAL						¥28.49

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Blank turning (face, OD, bore)	Skilled Machinist	¥71.49	1	4.93	90.0 %	¥6.53	¥25.57
Gear hobbing	Skilled Machinist	¥71.49	1	1.16	90.0 %	¥1.53	¥7.94
Chamfer and deburr	Semi-skilled Operator	¥49.77	1	0.43	90.0 %	¥0.39	¥1.50
Gear inspection	CMM / Quality	¥72.40	1	0.34	90.0	¥0.46	¥2.41

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
	Inspector				%		
TOTAL						¥8.92	¥37.42

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Hole / bore	Ø40.0 × 30 thru	1	Helical mill / bore	2.00	¥8.48	Yes
Face	8865 mm²	2	Face milling (facing)	2.62	¥11.09	Yes
TOTAL costed		3		4.62	¥19.56	Yes

\$5 — Machine Rate Buildup

CNC Lathe (2-axis) [mach-lathe-cnc] · Confidence: Low · UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥199095.02	¥62.22	4,000 hr/yr available
Maintenance	¥109502.26	¥34.22	
Energy	¥55085.58	¥17.21	
Floor Space	¥39819.00	¥12.44	
Indirect Support	¥109502.26	¥34.22	
Finance Cost	¥79638.01	¥24.89	
TOTAL — CNC Lathe (2-axis)	¥592642.14	¥185.20	Effective hours: 3200/yr

CNC Gear Hobber — small ("dm 4, "d Ø 200) [gear-hob-small] · Confidence: Low · REPRESENTATIVE — UK gear hobber class. Target ~£56/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥308597.29	¥96.44	4,000 hr/yr available
Maintenance	¥159276.02	¥49.77	
Energy	¥71611.25	¥22.38	
Floor Space	¥54751.13	¥17.11	
Indirect Support	¥149321.27	¥46.66	
Finance Cost	¥104524.89	¥32.66	
TOTAL — CNC Gear Hobber — small ("dm 4, "d Ø 200)	¥848081.84	¥265.03	Effective hours: 3200/yr

Gear Chamfer / Deburr Cell [gear-deburr] · Confidence: Low · REPRESENTATIVE — UK chamfer/deburr cell class. Target ~£26/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @82% util	Notes
Depreciation	¥129411.76	¥39.45	4,000 hr/yr available
Maintenance	¥69683.26	¥21.24	
Energy	¥33051.35	¥10.08	
Floor Space	¥39819.00	¥12.14	
Indirect Support	¥89592.76	¥27.31	
Finance Cost	¥44796.38	¥13.66	
TOTAL — Gear Chamfer / Deburr	¥406354.52	¥123.89	Effective hours: 3280/yr

CNC Gear Checker (profile / lead / pitch) [gear-checker] · Confidence: Low · REPRESENTATIVE — UK CNC gear metrology class (Klingelnberg/Gleason type). Target ~£56/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @65% util	Notes
Depreciation	¥288687.78	¥111.03	4,000 hr/yr available
Maintenance	¥109502.26	¥42.12	
Energy	¥27542.79	¥10.59	
Floor Space	¥59728.51	¥22.97	
Indirect Support	¥149321.27	¥57.43	
Finance Cost	¥74660.63	¥28.72	
TOTAL — CNC Gear Checker (profile / lead / pitch)	¥709443.24	¥272.86	Effective hours: 2600/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf
rawMaterial.directCost	3.9184	£	Pre-computed by commodity module	mat-steel4140	Medium
rawMaterial.consumablesCostPerPart	8.9486	£	Per-part consumable (core/wax/shell)	mat-steel4140	Medium
Blank turning (face, OD, bore).machineRatePerHr	20.4647	£/hr	UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-lathe-cnc	Low
Blank turning (face, OD, bore).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Gear hobbing.machineRatePerHr	29.2853	£/hr	REPRESENTATIVE — UK small CNC hobber class. Target ~£56/hr. Replace with plant data. Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-hob-small	Low
Gear hobbing.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Chamfer and deburr.machineRatePerHr	13.6897	£/hr	REPRESENTATIVE — UK chamfer/deburr cell class. Target ~£26/hr. Replace with plant data. Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-deburr	Low

Field	Value	Unit	Source / Reference	Rate ID	Conf.
Chamfer and deburr.labourRatePerHr	5.5000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-semiskilled	Low
Gear inspection.machineRatePerHr	30.1513	£/hr	REPRESENTATIVE — UK CNC gear metrology class (Klingelnberg/Gleason type). Target ~£56/hr. Replace with plant data. Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-checker	Low
Gear inspection.labourRatePerHr	8.0000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-inspector	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥193.14 ± 21.1%	¥155.11	¥189.86	¥236.47	wide	16.7%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 8 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread.

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.

Import duty and international freight (regional table is Ex-Works).

Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Direct material cost	¥35.46	-2.2%	+2.2%	¥8.58
Blank turning (face, OD, bore): Cycle Time	0.08hr	-1.6%	+1.6%	¥6.19
Blank turning (face, OD, bore): Machine Rate	¥185.20/hr	-1.2%	+1.2%	¥4.61
Overhead %	12%	-1.1%	+1.1%	¥4.06
Supplier Margin %	8%	-0.7%	+0.7%	¥2.86
Blank turning (face, OD, bore): Labour Rate	¥71.49/hr	-0.5%	+0.5%	¥1.95
Gear hobbing: Labour Rate	¥71.49/hr	-0.5%	+0.5%	¥1.95
Gear hobbing: Cycle Time	0.02hr	-0.5%	+0.5%	¥1.92

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥132.32	¥51.81	¥32.12	¥5.28	¥25.08	¥246.61	¥1.44	¥264.03	+37%
Germany (EUR)	¥136.29	¥54.40	¥51.91	¥5.54	¥30.91	¥279.05	¥1.65	¥298.65	+55%
France (EUR)	¥134.97	¥47.66	¥37.31	¥4.85	¥26.72	¥251.52	¥1.65	¥269.54	+40%
Spain (EUR)	¥132.32	¥45.59	¥23.52	¥4.64	¥22.17	¥228.25	¥1.65	¥244.88	+27%
Poland (PLN)	¥128.35	¥37.30	¥14.60	¥3.80	¥17.71	¥201.77	¥1.72	¥216.75	+12%
Turkey (TRY)	¥119.09	¥31.09	¥8.11	¥3.17	¥14.26	¥175.71	¥1.87	¥189.17	-2%
China (CNY)	¥116.44	¥28.49	¥8.92	¥2.90	¥18.81	¥175.57	¥2.08	¥193.14	Base
India (INR)	¥119.09	¥26.94	¥5.68	¥2.74	¥12.59	¥167.04	¥2.15	¥180.04	-7%
Mexico (MXN)	¥125.71	¥31.09	¥9.41	¥3.17	¥14.96	¥184.32	¥1.94	¥198.26	+3%
United States (USD)	¥132.32	¥44.04	¥42.18	¥4.49	¥23.99	¥247.01	¥1.79	¥264.75	+37%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

8.85 kgCO2e · 4.24 /kg · cradle-to-gate

Material	4.38 kgCO2e	Material factor	2.10 kg/kg (Steel)
Process	4.13 kgCO2e	Grid intensity	0.550 kg/kWh · 7.52 kWh
Logistics	0.33 kgCO2e	Total	8.85 kgCO2e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[CRITICAL] Material is the dominant cost driver

Finding	Raw material at 60.3% of total exceeds the gear benchmark ceiling of 50%. Material is controlling this part's economics.
Impact	High · up to 4% potential saving
Benchmark	Material %: yours 60.3% vs industry 28–50%
Action 1	Material utilisation at 100% — every +5% saves ~£0.64/part
Action 2	Re-cost the bought-out content at the target volume

§12 — Cost-Reduction Opportunities (ranked by saving)

7 opportunities · combined ¥21.63/part (~11%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Process & Automation	Focus cycle-time engineering on the governing operation before touching anything else; ask the supplier for its OEE and changeover data	One operation dominates — "Blank turning (face, OD, bore)" — "Blank turning (face, OD, bore)" carries 68% of the conversion cost across 4 operations. The part's cost rides on a single process step.	¥15.45	8.0%	Medium Term	Medium
2	Process & Automation	OEE Improvement Programme (TPM)	OEE at 80% vs 85% world-class target. Each 5% OEE gain reduces effective machine cost by 5%.	¥2.90	1.5%	Medium Term	Low
3	Material	Consumables Rationalisation (Cores, Patterns, Shell)	Per-part consumables are 70% of the material line. Core count, pattern life and shell recipe are all design-controllable.	¥11.59	6.0%	Medium Term	Medium
4	Design & Geometry	Design Review for DFM Compliance	Formal DFM review with manufacturing engineering to identify part features that increase cost without functional benefit.	¥9.66	5.0%	Quick Win	Low
5	Design & Geometry	Fix the burr at source (tooling discipline, gate redesign); where finishing must remain, tumble/vibratory beats manual	Manual finishing after the primary process — Deburring/fettling/rework appears as its own operation — a symptom of an upstream condition (tool wear, gating, die state) being paid for by hand, every part.	¥5.79	3.0%	Quick Win	Low
6	Commercial & Sourcing	Raw-Material Price Indexation Clause	With material at 60.3% of cost, the quote embeds the supplier's hedge against metal/resin volatility. An index-linked clause removes that risk premium from the piece price.	¥5.79	3.0%	Quick Win	Low
7	Commercial & Sourcing	Annual Volume Re-Commitment for Better Pricing	Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	¥5.79	3.0%	Quick Win	Low

§13 — Opportunity by Category

4 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Process & Automation	2	Focus cycle-time engineering on the governing operation before touching anything else; ask the supplier for its OEE and changeover data	¥15.45	¥18.35
Material	1	Consumables Rationalisation (Cores, Patterns, Shell)	¥11.59	¥11.59
Design & Geometry	2	Design Review for DFM Compliance	¥9.66	¥15.45
Commercial & Sourcing	2	Raw-Material Price Indexation Clause	¥5.79	¥11.59

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Raw-Material Price Indexation Clause
- Design Review for DFM Compliance
- Annual Volume Re-Commitment for Better Pricing